

Algorithms

Lecture 19:

Minimum Spanning Trees

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Review: BFS vs. DFS

	Breadth First Search (BFS)	Depth First Search (DFS)
Data Structure	use array and Queue	use array and Stack
Definition	a traversal approach in which we first walk through all nodes on the same level before moving on to the next level.	a traversal approach in which the traverse begins at the root node and proceeds through the nodes as far as possible until we reach the node with no unvisited nearby nodes.
Visiting of Siblings/Children	siblings are visited before the children	children are visited before the siblings
Time Complexity	$O(m + n)$ when Adjacency List is used, and $O(n^2)$ when Adjacency Matrix is used, where n stands for # of vertices and m stands for # of edges.	
Applications	find a shortest path in an unweighted graph	acyclic graphs, topological sort

Minimum Spanning Trees

Weighted Graph Algorithms

- Beyond DFS/BFS exists an alternate universe of algorithms for edge-weighted graphs.
- Our adjacency list representation quietly supported these graphs:

```
typedef struct edgenode {
    int y;                /* adjacency info */
    int weight;           /* edge weight, if any */
    struct edgenode *next; /* next edge in list */
} edgenode;

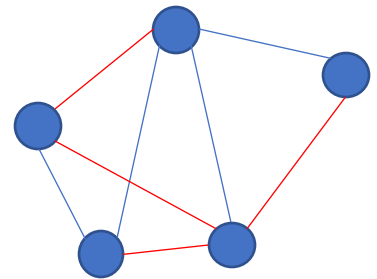
typedef struct {
    edgenode *edges[n]; /* adjacency info */
    int degree[n];      /* outdegree of each vertex */
    int nvertices;      /* number of vertices in the graph */
    int nedges;         /* number of edges in the graph */
    int directed;       /* is the graph directed? */
} graph;
```

Spanning Trees

- A tree is a connected graph with no cycles.
- A **spanning tree** is a subgraph of G that covers all the vertices of G and is a tree.

- Discussion:

- Do the blue edges make a spanning tree?
 - Do the red edges make a spanning tree?
- What graph has the most spanning trees?
- What graph has the fewest spanning trees?
- Give an algorithm to find a spanning tree of a graph.

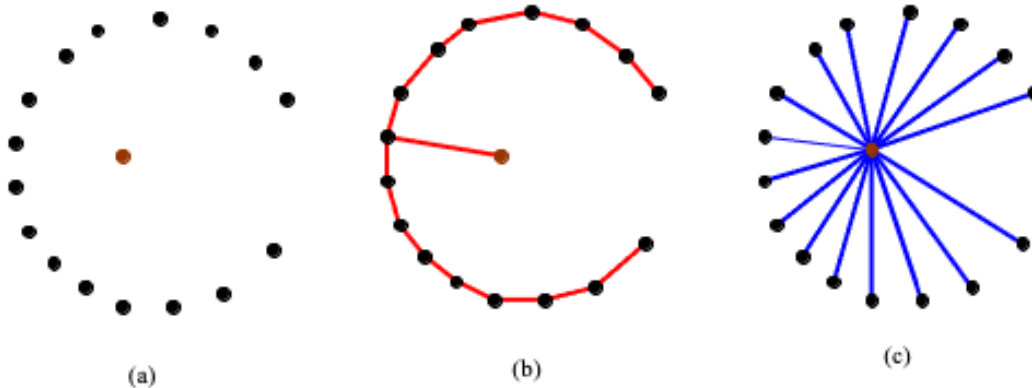


Minimum Spanning Trees

- A **spanning tree** is a subgraph of G that covers all the vertices of G and is a tree.
- A **minimum spanning tree (MST)** of a weighted graph G is the spanning tree of G whose edges sum to minimum weight.
- There can be more than one minimum spanning tree in a graph → consider a graph with identical weight edges.
- Finding an MST is a fundamental problem on weighted graphs.

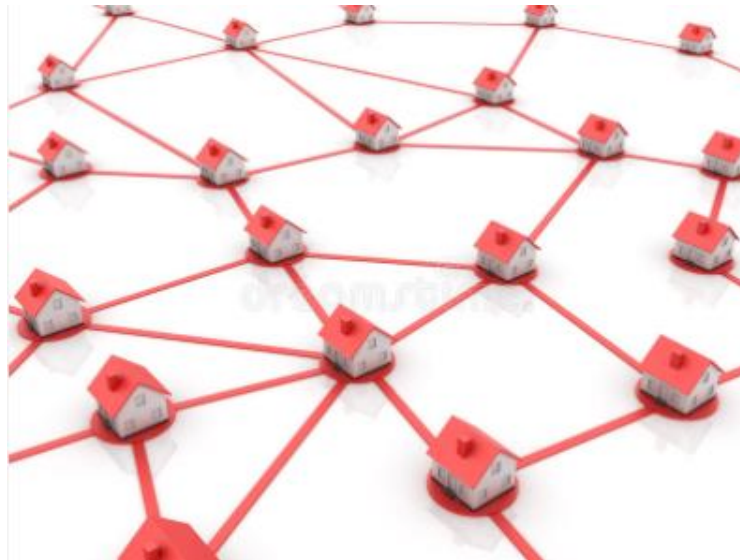
Example of Minimum Spanning Tree

- (a) Edges are not showing on the graph, but every point is connected to every other point, and the weight is the distance between them.
- (b) and (c) are spanning trees of (a)
- Which one of (b) and (c) are the one with less weight.



Applications of Minimum Spanning Trees

- Minimum spanning trees are useful in constructing networks, by describing the way to connect a set of sites using the smallest total amount of wire.



Algorithms to find MSTs

- Prim's Algorithm
- Kruskal's Algorithm